

IKEv1 aggressive mode (edge case)

Analysed 29/09/2026, 02:44:59 · 1 tunnel · model bundle v1 (a88b2a4) · contract antardrishti.result/1



4 checks failed or raised a warning across 1 tunnel.

5 checks could not be decided from this capture (for example, PFS needs a rekey in the capture); they are listed as not determinable, never guessed.

Top findings

- High

tunnel 0 · IKEv1 aggressive mode

IKEv1 aggressive mode sends the identity in the clear and, with a pre-shared key, a hash an attacker can crack offline
- Medium

tunnel 0 · IKE version

The tunnel uses IKEv1, which NIST SP 800-77r1 recommends against; use IKEv2
- Medium

tunnel 0 · Traffic metadata exposure

A passive observer can tell the traffic inside apart: icmp (100% of bytes, presence 99%)
- Low

tunnel 0 · Peer authentication

IKEv1 authenticates with a pre-shared key; its strength cannot be seen

Traffic overview

Tunnel	Handshake	What it carries (share of bytes)	Risk
#0	full	ICMP 100%	39 Medium

Recommendations

1. **IKEv1 aggressive mode:** Disable aggressive mode (or move to IKEv2).
2. **IKE version:** Migrate the tunnel to IKEv2.
3. **Traffic metadata exposure:** Enable TFC padding (RFC 4303 section 2.7) or dummy traffic where hiding activity matters.
4. **Peer authentication:** Prefer certificate (RSA-PSS or ECDSA) authentication; if a PSK is used, make it long and random.

Passive analysis of the outer capture only: no traffic was decrypted and no keys were used. Facts marked "inferred" come from models trained on the Antardrishti dataset; their confidence is calibrated, and findings below 80% confidence are worded "likely".